Analysis III
Home Work 4

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Exercise 1 (Fubini's Theorem, 2+3+1 P.) Define $f : [0;1]^2 \to \mathbb{R}$ by $f(0,0) = 0$ and

$$f(x,y) = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

for $(x,y) \neq (0,0)$.

(a) Prove that, for $(x,y) \in (0;1)^2$,

$$f(x,y) = \frac{\partial}{\partial x}\frac{x}{x^2+y^2} = -\frac{\partial}{\partial y}\frac{y}{x^2+y^2}.$$

(b) Compute the integrals

$$\int_0^1 \left[\int_0^1 f(x,y)\,dx\right] dy \qquad \text{and} \qquad \int_0^1 \left[\int_0^1 f(x,y)\,dy\right] dx.$$

(c) Is $f \in L^1([0;1]^2;\mathbb{R})$? Justify.

Answer:

(a) We have that:

$$\begin{aligned}\frac{\partial}{\partial x} x(x^2+y^2)^{-1} &= (x^2+y^2)^{-1} + x\left[\frac{-1}{(x^2+y^2)^2}\cdot 2x\right] \\ &= \frac{1}{(x^2+y^2)^2}\left[x^2+y^2-2x^2\right] \\ &= \frac{y^2-x^2}{(x^2+y^2)^2}.\end{aligned}$$

Similarly:

$$\frac{\partial}{\partial y} y(x^2+y^2)^{-1} = \frac{x^2-y^2}{(x^2+y^2)^2}.$$

i.e.

$$\begin{aligned}\frac{\partial}{\partial x} x(x^2+y^2)^{-1} &= -\frac{\partial}{\partial y} y(x^2+y^2)^{-1} \\ &= f(x,y)\end{aligned}$$

Exercise 2 (Application of the MCT, 2+2 P.) Let $f \in L^1(\mathbb{R}^n;\mathbb{R})$ and denote by $B_\varepsilon(x)$ the ball of radius $\varepsilon > 0$ around $x \in \mathbb{R}^n$. We want to prove that

$$\lim_{\varepsilon \to 0^+} \int_{B_\varepsilon(x)} f(y)\, dy = 0.$$

(a) Use the Monotone Convergence Theorem to prove the assertion under the additional assumption that $f \geq 0$. *Hint:* You may use without proof that $f\chi_B \in L^1(\mathbb{R}^n;\mathbb{R})$ for any ball $B \subset \mathbb{R}^n$.

(b) Prove the assertion without the additional assumption by reducing the general problem to the situation in (a). *Hint:* Sheet 3, Exercise 1 might be useful.

Remark. It is in general a good strategy to prove statements first for non-negative functions and then generalize.

Answer: